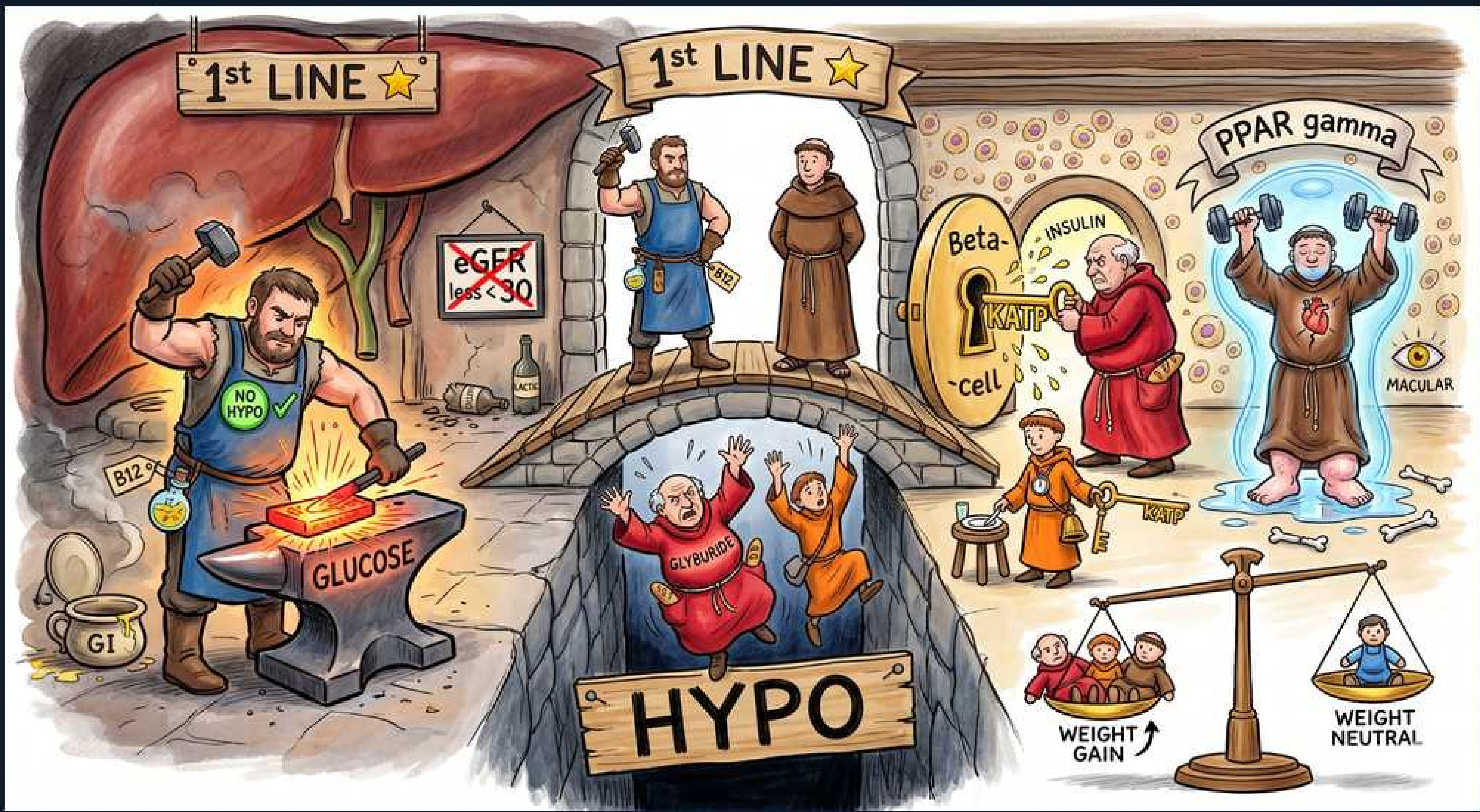
DM — Scene 1 — Diabetes Pharmacology: The Liver-Pancreas Workshop



1. Metformin blacksmith (1st line)

WHAT YOU SEE

Blacksmith at liver anvil hammering GLUCOSE, 1ST LINE star

WHAT IT ENCODES

Metformin is the typical first-line pharmacotherapy for T2DM. Its primary mechanism is reducing hepatic gluconeogenesis (it activates AMPK), with secondary improvement in peripheral insulin sensitivity. It lowers A1c ~1-1.5%, is weight-neutral/mildly weight-losing, inexpensive, and does not cause hypoglycemia as monotherapy. Start low and titrate to limit GI effects.

BOARD TRAP

Wrong answer: 'start a sulfonylurea or insulin as routine first-line therapy in uncomplicated new T2DM.' Discriminator — metformin is foundational first-line (efficacy, weight-neutral, no hypoglycemia, low cost); secretagogues/insulin are not the default initial agents unless the patient is markedly symptomatic/catabolic.

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2. No hypo badge

WHAT YOU SEE

Blue NO HYPO checkmark shield

WHAT IT ENCODES

Metformin does not cause hypoglycemia when used alone because it works independently of insulin secretion (it suppresses hepatic glucose output rather than stimulating insulin release). Hypoglycemia can still occur if it is combined with a sulfonylurea or insulin.

BOARD TRAP

Wrong answer: 'attribute a metformin-monotherapy patient's hypoglycemia to the metformin.' Discriminator — metformin alone does not cause hypoglycemia; look for a co-prescribed secretagogue (sulfonylurea/meglitinide) or insulin as the true cause.

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3. eGFR < 30 contraindication

WHAT YOU SEE

Road sign eGFR less than 30

WHAT IT ENCODES

Stop/avoid metformin when eGFR <30 mL/min/1.73m² because of lactic acidosis risk from drug accumulation; use caution and do not initiate when eGFR is 30-45 (reduce dose / monitor). Also hold around iodinated contrast in at-risk patients and during acute kidney injury.

BOARD TRAP

Wrong answer: 'continue full-dose metformin in a patient with eGFR 22 (advanced CKD).' Discriminator — eGFR <30 is a contraindication due to lactic acidosis risk; the drug must be stopped, not merely monitored, at this level of renal function.

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4. B12 deficiency

WHAT YOU SEE

B12 pot/bottle by blacksmith

WHAT IT ENCODES

Long-term metformin use causes vitamin B12 deficiency by impairing ileal B12 absorption. This can produce macrocytic anemia and/or peripheral neuropathy — the latter easily mistaken for diabetic neuropathy. Periodically check B12 in long-term users, especially with neuropathy or anemia.

BOARD TRAP

Wrong answer: 'assume new peripheral neuropathy in a long-term metformin user is purely diabetic neuropathy.' Discriminator — metformin-induced B12 deficiency can cause/worsen neuropathy; check a B12 level before attributing it solely to diabetes.

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5. GI side effect

WHAT YOU SEE

A toilet icon labelled GI tucked at the blacksmith's feet, bottom-left — the toilet is metformin's diarrhea/GI upset, its most common side effect and #1 reason for intolerance.

WHAT IT ENCODES

GI upset (diarrhea, nausea, abdominal discomfort, metallic taste) is metformin's most common adverse effect and the leading reason for intolerance. Mitigate by starting low, titrating slowly, taking with food, or switching to extended-release (XR), which is better tolerated.

BOARD TRAP

Wrong answer: 'abandon metformin entirely at the first sign of GI upset.' Discriminator — GI effects are usually transient and dose-related; slow titration, dosing with food, or extended-release formulation often resolves them before considering an alternative agent.

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6. Lactic acidosis (alcohol/contrast)

WHAT YOU SEE

Bottles/contrast near liver, ruined background

WHAT IT ENCODES

Metformin-associated lactic acidosis is rare but serious; risk rises with conditions causing lactate accumulation or drug retention — renal failure, hypoxia/sepsis, heart failure, hepatic dysfunction, heavy alcohol use, and iodinated contrast. Hold metformin around contrast administration and in any acute illness threatening renal perfusion.

BOARD TRAP

Wrong answer: 'give IV contrast for the CT and continue metformin unchanged in a patient with borderline renal function.' Discriminator — contrast plus impaired renal clearance can precipitate metformin accumulation and lactic acidosis; hold metformin peri-contrast and reassess renal function before resuming.

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7. Sulfonylurea monk + KATP key

WHAT YOU SEE

Monk closing Beta-cell KATP channel with key, INSULIN released, 1ST LINE bridge

WHAT IT ENCODES

Sulfonylureas (glipizide, glyburide, glimepiride) are insulin SECRETAGOGUES: they close the beta-cell ATP-sensitive potassium (KATP) channel, causing membrane depolarization, calcium influx, and insulin release — independent of glucose level. They lower A1c ~1-1.5% but require functioning beta cells. Meglitinides (repaglinide, nateglinide) act on the same channel with faster, shorter action for postprandial control.

BOARD TRAP

Wrong answer: 'a sulfonylurea works by reducing insulin resistance.' Discriminator — sulfonylureas STIMULATE insulin secretion (secretagogues); the insulin-sensitizers are metformin and the TZDs.

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8. HYPO pit (sulfonylurea risk)

WHAT YOU SEE

Monks falling into HYPO pit under bridge

WHAT IT ENCODES

Because sulfonylureas drive insulin release independent of glucose, hypoglycemia is their hallmark adverse effect — and it can be prolonged and severe. Risk is greatest in the elderly and in renal impairment; glyburide is the worst offender (long-acting, renally cleared active metabolites) and should be avoided in older/CKD patients in favor of glipizide.

BOARD TRAP

Wrong answer: 'a sulfonylurea is a hypoglycemia-safe choice for an elderly patient with CKD.' Discriminator — sulfonylureas (especially glyburide) are a leading cause of severe, prolonged hypoglycemia in the elderly/CKD; safer alternatives or short-acting glipizide are preferred.

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9. Weight gain scale (SU)

WHAT YOU SEE

Scale tipping WEIGHT GAIN under sulfonylureas

WHAT IT ENCODES

Sulfonylureas cause weight gain (from increased insulin secretion and anabolic effect), which is a disadvantage in overweight/obese T2DM patients. Insulin and TZDs likewise add weight, whereas metformin and SGLT2 inhibitors are weight-neutral/losing and GLP-1 RAs cause weight loss.

BOARD TRAP

Wrong answer: 'choose a sulfonylurea for an obese T2DM patient who needs to lose weight.' Discriminator — sulfonylureas promote weight GAIN; in obesity, prefer a weight-losing agent (GLP-1 RA) or weight-neutral agent (metformin, SGLT2 inhibitor).

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10. Metformin weight neutral scale

WHAT YOU SEE

Scale WEIGHT NEUTRAL

WHAT IT ENCODES

Metformin is weight-neutral and often mildly weight-losing, one reason it is favored as the foundational agent — particularly in overweight/obese T2DM. This contrasts with the weight-gaining sulfonylureas, TZDs, and insulin.

BOARD TRAP

Wrong answer: 'avoid metformin in an obese diabetic because oral diabetes drugs all cause weight gain.' Discriminator — metformin is weight-neutral/mildly weight-losing; the weight-gaining oral agents are sulfonylureas and TZDs, not metformin.

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11. TZD / PPAR-gamma figure

WHAT YOU SEE

Robed figure lifting weights, PPAR GAMMA banner

WHAT IT ENCODES

Thiazolidinediones (TZDs; pioglitazone, rosiglitazone) are PPAR-gamma agonists that enhance peripheral insulin sensitivity in muscle/adipose tissue (insulin sensitizers, not secretagogues). They lower A1c ~1%, do not cause hypoglycemia alone, and have a slow onset (weeks). Pioglitazone is the preferred TZD and may benefit NASH; rosiglitazone has had CV concerns.

BOARD TRAP

Wrong answer: 'a TZD causes hypoglycemia like a sulfonylurea.' Discriminator — TZDs are insulin sensitizers acting via PPAR-gamma and do not cause hypoglycemia as monotherapy; the secretagogues (sulfonylureas/meglitinides) and insulin do.

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12. TZD edema / heart failure

WHAT YOU SEE

Swollen feet of TZD figure

WHAT IT ENCODES

TZDs cause sodium/fluid retention leading to peripheral edema and weight gain, and they can precipitate or worsen heart failure — TZDs are contraindicated in NYHA class III-IV HF. The fluid retention can also potentiate edema when combined with insulin.

BOARD TRAP

Wrong answer: 'add pioglitazone for an HFrEF patient with poorly controlled diabetes.' Discriminator — TZD-induced fluid retention worsens heart failure (contraindicated in symptomatic HF); choose an SGLT2 inhibitor, which is HF-protective, instead.

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13. TZD macular edema (eye)

WHAT YOU SEE

A single eye symbol tagged MACULAR beside the weight-lifting TZD figure — the eye is drug-induced macular edema (blurred vision) from TZD fluid retention.

WHAT IT ENCODES

TZDs can cause macular edema (related to their fluid-retention effect), presenting with blurred or decreased vision. Consider drug-induced macular edema in a TZD-treated patient with new visual changes and refer for ophthalmologic evaluation.

BOARD TRAP

Wrong answer: 'attribute new macular edema in a TZD-treated patient solely to diabetic retinopathy.' Discriminator — TZDs are a recognized cause of macular edema; the drug should be considered (and possibly stopped) rather than assuming it is purely diabetic retinopathy.

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14. TZD bone fracture

WHAT YOU SEE

Broken bone at TZD figure feet

WHAT IT ENCODES

TZDs increase the risk of bone fractures (especially distal limb fractures in postmenopausal women) by promoting adipogenesis over osteoblastogenesis, reducing bone formation. Weigh this against benefits in patients at high fracture risk/osteoporosis.

BOARD TRAP

Wrong answer: 'pioglitazone has no skeletal effects, so it's a fine choice for an osteoporotic postmenopausal woman.' Discriminator — TZDs increase fracture risk and accelerate bone loss; avoid them in patients with osteoporosis or high fracture risk.
